1. What is a Data Structure?

Answer: A data structure is a way of organizing and storing data so that it can be accessed and modified efficiently.

2. What is an Algorithm?

Answer: An algorithm is a step-by-step procedure used to solve a particular problem.

3. What is an Array?

Answer: An array stores multiple elements of the same type in a continuous memory location. Elements can be accessed using an index.

4. What is a Linked List?

Answer: A linked list is a collection of nodes where each node contains data and a link to the next node.

5. What is a Stack?

Answer: A stack is a linear data structure that follows LIFO – Last In, First Out. Example: the undo operation in an application.

6. What is a Queue?

Answer: A queue follows FIFO – First In, First Out. The element inserted first is removed first. Example: people waiting in a line.

7. What is a Tree?

Answer: A tree is a hierarchical data structure consisting of nodes connected by edges. It has a root node and child nodes.

8. What is a Graph?

Answer: A graph is a data structure consisting of vertices (nodes) and edges that represent relationships between them.

9. What is Searching?

Answer: Searching is the process of finding a particular element in a collection of data. Common methods include linear search and binary search.

10. What is Sorting?

Answer: Sorting is the process of arranging data in a particular order, such as ascending or descending order. Examples include bubble sort, selection sort, and merge sort.